

Green Hydrogen Microgrids

Concepts, Definitions, Classical Control and Reinforcement-Learning
Integration

A technical primer built around a DC islanded microgrid that produces green hydrogen from a solar array and a battery, with an electrolyzer as the load. The plant is modelled in MATLAB / Simulink / Simscape and serves as the environment for reinforcement-learning based supervisory control. All curves are derived from the model parameters in **GreenHydrogenMicrogridData.m**.

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Generated for the **GreenHydrogenMicrogridExample** repository · MATLAB/Simulink/Simscape
plant © The MathWorks, Inc. · 13 figures included.

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1 · Introduction and Reading Guide

Decarbonising energy means electrifying everything we can and finding a clean carrier for everything we cannot electrify directly. **Green hydrogen** — hydrogen split from water using renewable electricity — is the leading candidate for that second category: long-duration storage, heavy industry, and hard-to-electrify transport. A **microgrid** is the natural place to produce it: a small, self-contained power system that can marry an intermittent renewable source to a flexible electrolyzer load and a battery buffer.

This document is a self-contained primer that walks from first principles to learned control, organised around one concrete system — the **GreenHydrogenMicrogridExample** plant. It explains *what* green hydrogen and microgrids are, *how* they relate, *how* such a system is classically controlled (with the pros and cons of each method), and *how* reinforcement learning (RL) can be integrated to do better. Every curve in this report is generated from the same parameters the Simscape model uses, so the numbers line up with the code in the repository.

How to read this report. Sections 2–3 define the two halves (hydrogen, microgrids). Section 4 couples them and states the control problem. Section 5 surveys classical controllers with explicit pros/cons tables. Section 6 introduces RL and shows how it plugs into the very same plant. If you only want the punch-line, jump to the comparison in Fig. 9 and the KPI summary in Section 7.

2 · Green Hydrogen — Concepts and Definitions

2.1 Hydrogen as an energy carrier

Hydrogen (H_2) is not an energy *source* — there are no hydrogen wells. It is an energy *carrier*: a way to store and move energy that was generated elsewhere. Its appeal is threefold. First, **gravimetric energy density** is exceptional — about 120–142 MJ/kg (33–39.4 kWh/kg), roughly three times that of diesel by mass. Second, **using** it (in a fuel cell or burner) emits only water. Third, it **decouples** the time and place of renewable generation from the time and place of demand: sunshine at noon can become a truckload of fuel, or electricity at midnight.

The catch is volumetric density and round-trip efficiency. As a gas at ambient pressure hydrogen is extremely bulky, so it must be compressed, liquefied, or chemically bound; and every conversion (electricity $\rightarrow \text{H}_2 \rightarrow$ electricity) loses energy. Understanding those losses — quantified in Fig. 13 — is central to controlling a hydrogen microgrid well.

2.2 Water electrolysis: the core reaction

Green hydrogen is made by **water electrolysis**: passing a direct current through water to split it into hydrogen and oxygen. The overall reaction is



The thermodynamic minimum cell voltage at standard conditions is the **reversible voltage** $E_{\text{rev}} \approx 1.23 \text{ V}$ (1.48 V if the reaction's heat demand is also supplied electrically — the *thermoneutral* voltage). Real cells run above this to overcome activation, ohmic and mass-transport losses. The hydrogen production rate follows **Faraday's law**: it is proportional to the current (number of electrons per second), so to first order *more current = more hydrogen*, while the *voltage* at that current sets the efficiency. This current/voltage split is exactly what the polarisation curve in Fig. 2 captures, and why the controller's job is to keep the electrolyzer in an efficient operating band.

2.3 The hydrogen colour taxonomy

Hydrogen is colourless; the “colours” are a shorthand for the **carbon intensity of how it was produced**. This is the single most important distinction in the hydrogen economy, because most hydrogen made today is anything but clean.

- **Grey** — from steam-methane reforming (SMR) of natural gas; ~9–10 kg CO_2 per kg H_2 . The incumbent, and cheap.
- **Brown / Black** — from coal gasification; the most carbon-intensive pathway (~20 kg CO_2 /kg).
- **Blue** — grey hydrogen with carbon capture and storage (CCS); cuts but does not eliminate emissions.
- **Yellow** — electrolysis powered by grid electricity; only as clean as the grid mix behind it.
- **Green** — electrolysis powered by dedicated renewables (solar/wind/hydro); near-zero operational CO_2 . **This is what our microgrid produces.**

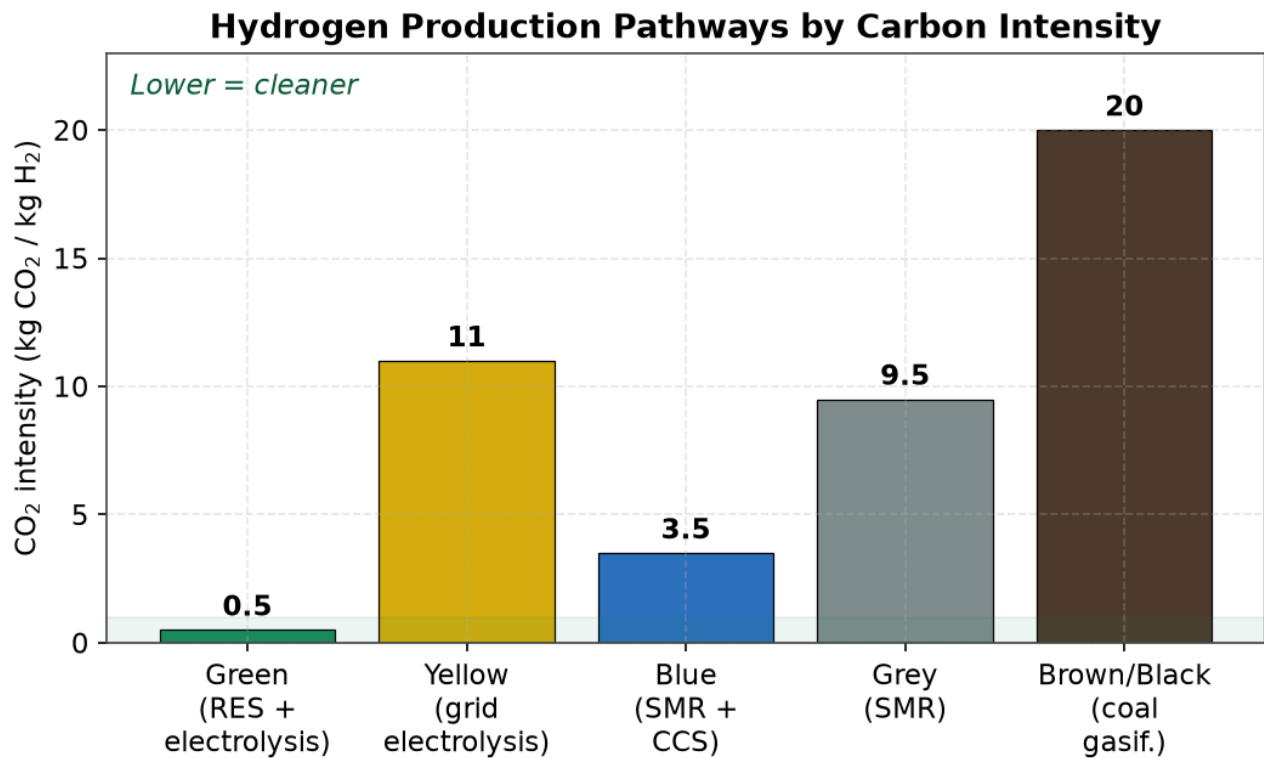


Figure 1. Representative life-cycle CO₂ intensity of the main hydrogen production pathways. Green hydrogen (renewables + electrolysis) sits near zero; grey and brown hydrogen — today's dominant routes — are an order of magnitude higher.

2.4 Electrolyzer technologies

Four electrolyzer families dominate, differing in electrolyte, operating temperature and dynamics. The dynamics matter for control: an electrolyzer that can ramp quickly is a far more useful partner for intermittent solar.

Technology	Pros	Cons
Alkaline (AEL)	Mature, low-cost, durable, no precious metals; large installed base.	Slower dynamics, lower current density, limited low-load turndown; liquid KOH electrolyte management.
PEM	Fast ramping, high current density, compact, wide turndown — ideal for variable renewables; high-purity H ₂ .	Uses platinum-group catalysts and costly membranes; acidic, more expensive than alkaline.
Solid-oxide (SOEC)	Highest efficiency; can use waste heat; reversible (also runs as a fuel cell).	Very high temperature (700–850 °C); thermal cycling degrades ceramics; still maturing.
Anion-exchange (AEM)	Aims to combine PEM dynamics with alkaline (non-precious) materials; low cost potential.	Earliest stage of commercialisation; membrane durability still improving.

The example plant models a stack of **100 cells** with **3 electrode pairs**, a 0.25 Ω internal resistance and a temperature-dependent efficiency — behaviour most consistent with a PEM/alkaline-style unit driven through a DC/DC converter.

2.5 The polarisation curve

The **polarisation (V–I) curve** is the electrolyzer's signature: cell voltage as a function of current density. It decomposes into three regions. At low current, **activation overpotential** dominates (the energy to start

the reaction). In the broad middle, voltage rises **linearly** with current — the **ohmic region** set by membrane and electrode resistance. At very high current, **mass-transport** limits (reactant supply, bubble blanketing) make voltage shoot up. Because power = $V \times I$ and efficiency $\propto E_{\text{rev}}/V$, the sweet spot is the ohmic region: enough current for good throughput, before losses balloon. A good supervisor keeps the stack here.

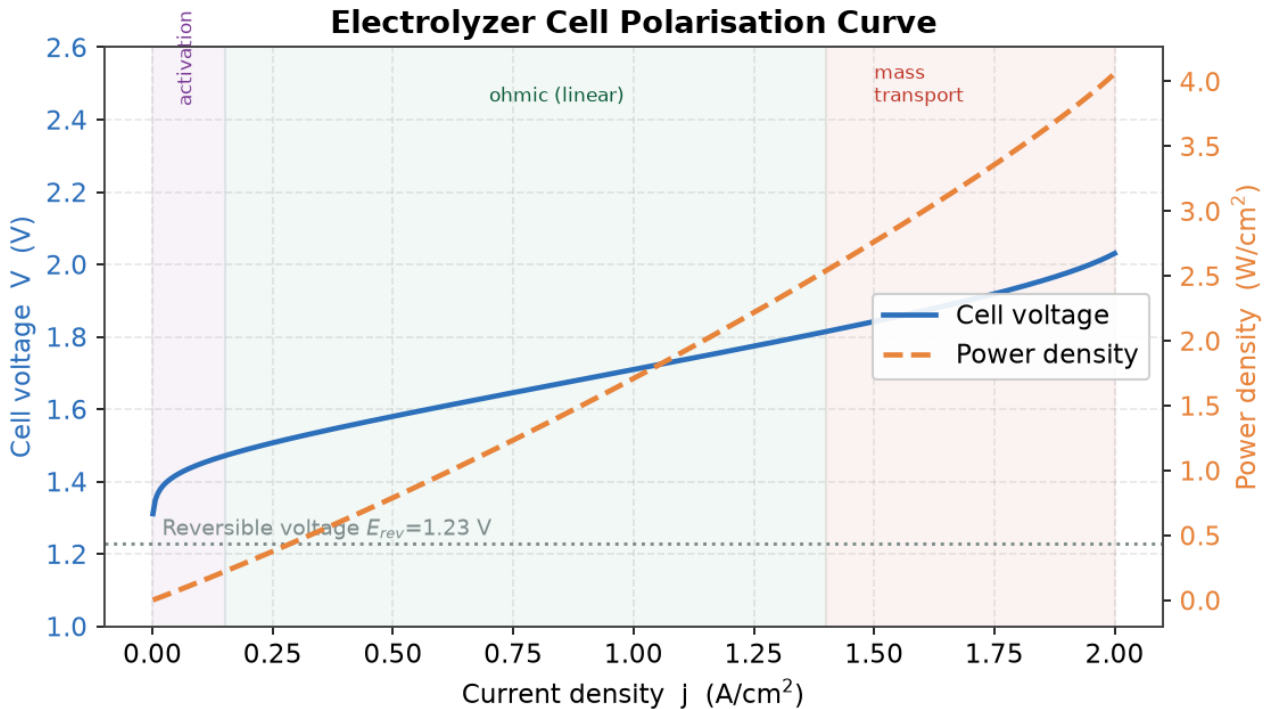


Figure 2. Electrolyzer single-cell polarisation curve (blue) and power density (orange). Shaded bands mark the activation, ohmic (linear) and mass-transport regimes. The reversible voltage (1.23 V) is the thermodynamic floor; all voltage above it is loss/heat.

2.6 Efficiency, HHV/LHV and Faraday efficiency

Three efficiency notions recur. **Voltage (energy) efficiency** = $E_{\text{thermoneutral}}/V_{\text{cell}}$ — how close the cell runs to its ideal voltage. **Faraday (current) efficiency** — the fraction of current that actually makes hydrogen rather than parasitic reactions or cross-over (usually >95% in healthy stacks). **System efficiency** folds in the converter, balance-of-plant and compression. Energy content is quoted against the **higher heating value (HHV ≈ 39.4 kWh/kg)** or **lower heating value (LHV ≈ 33.3 kWh/kg)**; this report uses HHV, matching the model's hydrogen-mass accounting.

Crucially, electrolyzer efficiency **rises with temperature**: hotter electrolyte lowers the activation and ohmic overpotentials, so the same current is delivered at a lower voltage. The model encodes this directly as an efficiency-vs-temperature lookup, shown in Fig. 3. This is a control lever: warming the stack (within limits) buys efficiency, but the heater draws power — a trade-off an RL agent can learn to exploit.

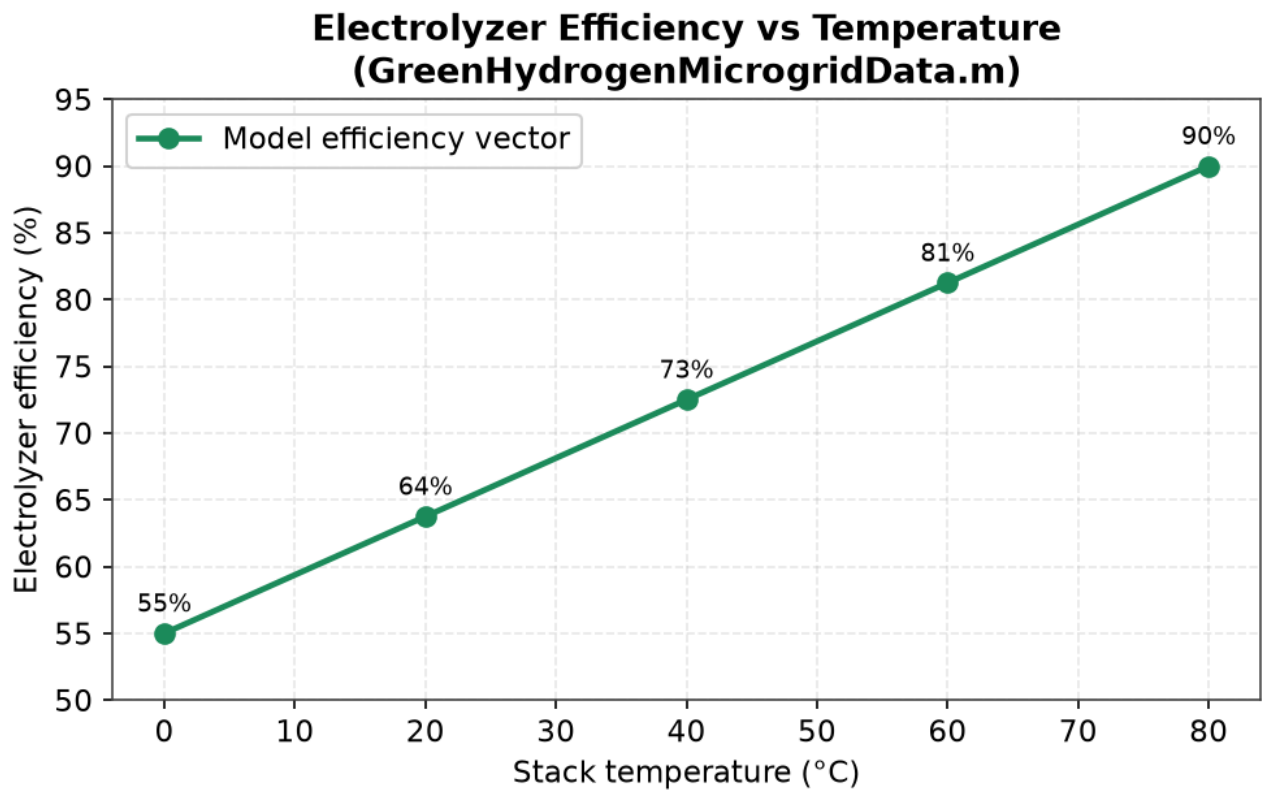


Figure 3. Electrolyzer efficiency vs stack temperature, taken directly from `GreenHydrogenMicrogridData.m` (Electrolyzer.Efficiency_vect rising 55% → 90% across 0–80 °C). Higher temperature = higher efficiency.

2.7 Power-to-X and sector coupling

Producing hydrogen from surplus renewable power is one instance of **Power-to-X** (PtX): converting electricity into a storable, transportable vector. Hydrogen then **couples sectors** that electricity alone struggles to reach — it can be re-electrified in a fuel cell or turbine, used as industrial feedstock (ammonia, steel, refining), or fuel mobility. This is why a hydrogen microgrid is more than a battery: it turns a local electricity surplus into a product with many downstream uses and effectively unlimited storage duration.

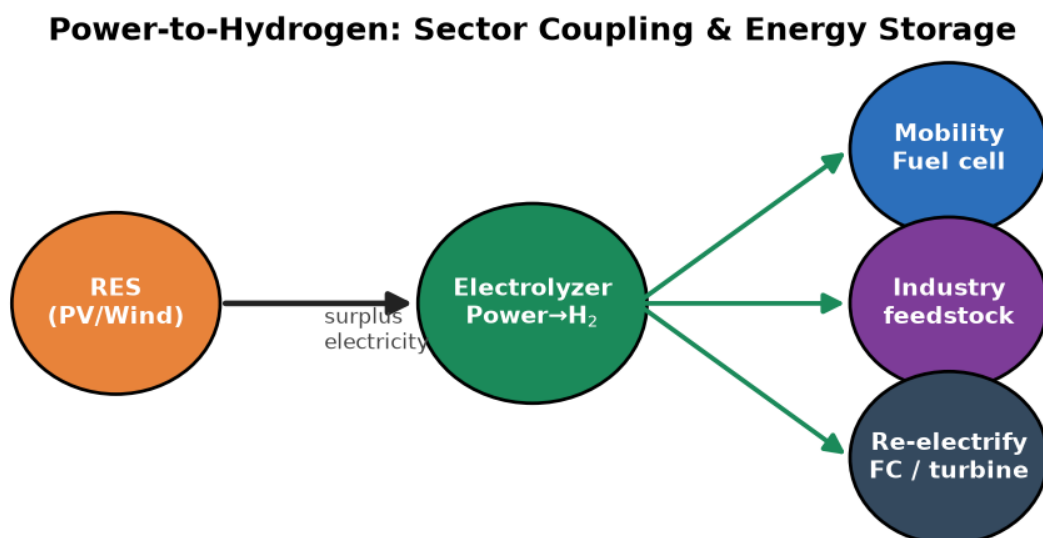


Figure 12. Power-to-Hydrogen as sector coupling. Surplus renewable electricity drives the electrolyzer; the resulting hydrogen feeds mobility, industry, or re-electrification — decoupling generation from end-use in both time and sector.

2.8 Energy flow and losses

Every stage from electricity to stored hydrogen costs energy. A representative chain for this kind of plant: the DC/DC converter is >95% efficient; electrolysis is the big loss (the stack runs at ~65–75% on an HHV basis at typical loads); and storage/compression shaves off a little more. The result, Fig. 13, is why round-trip power→H₂→power efficiency is modest (~30–40%) and why hydrogen is reserved for jobs batteries cannot do — long-duration and non-electric end-uses. A controller that keeps the electrolyzer in its efficient band and avoids needless conversions directly moves these numbers.

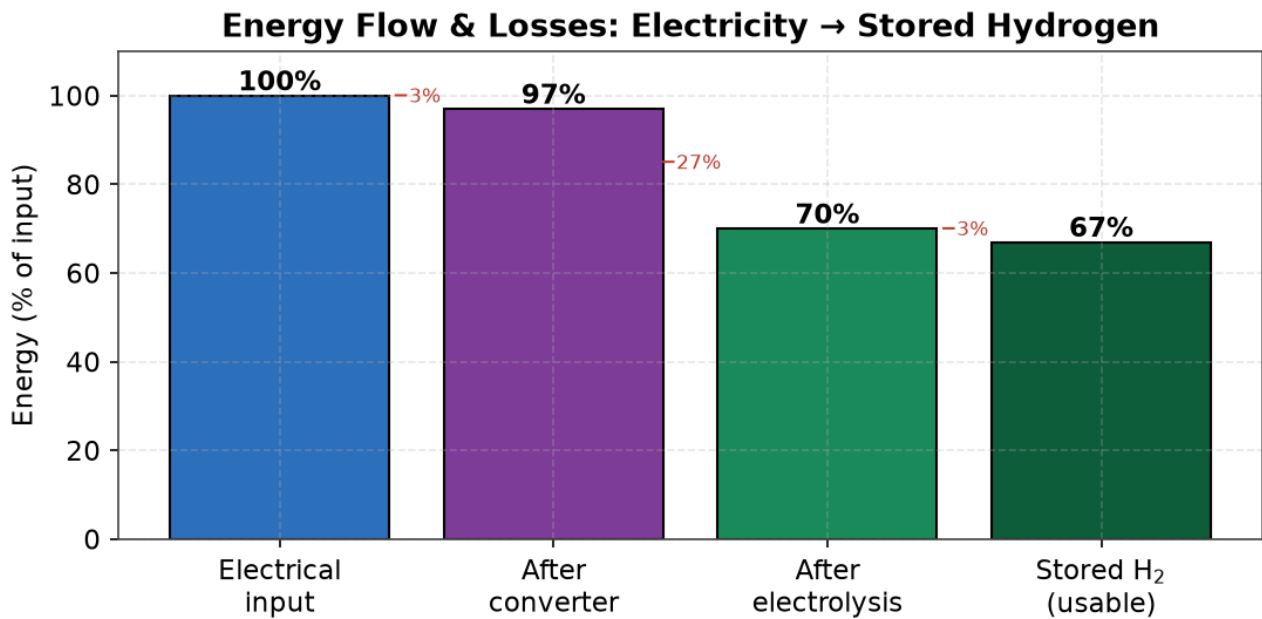


Figure 13. Energy cascade from electrical input to usable stored hydrogen. The dominant loss is the electrolysis step itself; converter and storage losses are comparatively small.

3 · Microgrids — Concepts and Definitions

3.1 What is a microgrid?

A **microgrid** is a localised group of electricity sources and loads with clearly defined electrical boundaries that can operate either connected to the wider grid or **islanded** (autonomously). The U.S. Department of Energy's working definition captures the essence: a microgrid acts as a *single controllable entity* with respect to the grid, and can disconnect to run on its own. Microgrids appear on campuses, remote communities, islands, military bases and industrial sites — anywhere reliability, resilience or renewable integration matters.

3.2 AC vs DC, grid-connected vs islanded

- **Grid-connected** microgrids lean on the main grid for backup and frequency/voltage reference; the grid absorbs imbalance.
- **Islanded** microgrids must balance generation and load entirely on their own — there is no external buffer, so control and storage are critical. *Our example is a DC islanded microgrid.*
- **AC microgrids** are the traditional choice and interoperate with existing equipment, but need synchronisation and reactive-power management.
- **DC microgrids** suit PV, batteries and electrolyzers — all natively DC — avoiding several conversion stages, with no frequency or reactive power to manage. The price is mature DC protection and standards.

Why DC here? Solar arrays, batteries and electrolyzers are all DC devices. Tying them to a common **DC bus** removes redundant DC–AC–DC conversions, raising efficiency and simplifying control — which is exactly the topology of this example.

3.3 Building blocks

A microgrid is assembled from a handful of functional blocks:

- **Distributed generation (DG):** renewable (PV, wind) and/or dispatchable (gensets, fuel cells). Here: a ~210 kW **solar array**.
- **Energy storage:** smooths the mismatch between generation and load. Here: a **battery** (240 V nominal, 50 Ah-class) — and, uniquely, the **hydrogen tank** acts as long-duration storage.
- **Loads:** fixed or flexible. Here the electrolyzer is a large **flexible load** — it can be throttled, which is what makes smart control valuable.
- **Power-electronic converters:** DC/DC and DC/AC interfaces with their own efficiency maps (Fig. 4, left).
- **Supervisory controller / EMS:** the “brain” that decides power flows. **This is what classical methods and RL compete to implement.**

3.4 The control hierarchy

Microgrid control is organised as a three-level **hierarchy** separated by time-scale — a structure that keeps fast stability concerns away from slow economic ones:

- **Primary (milliseconds–seconds):** local voltage/current loops, droop control and converter PWM. Keeps the bus stable instant-to-instant.

- **Secondary (seconds–minutes):** restores voltage/SOC set-points and balances power among units, correcting the small deviations primary control leaves behind.
- **Tertiary / Energy-Management System (minutes–hours):** economic dispatch, scheduling and — here — **hydrogen production planning**. This slow, decision-rich layer is where classical rules/MPC live and **where the RL agent is inserted**.

Microgrid Control Hierarchy (RL targets the EMS layer)

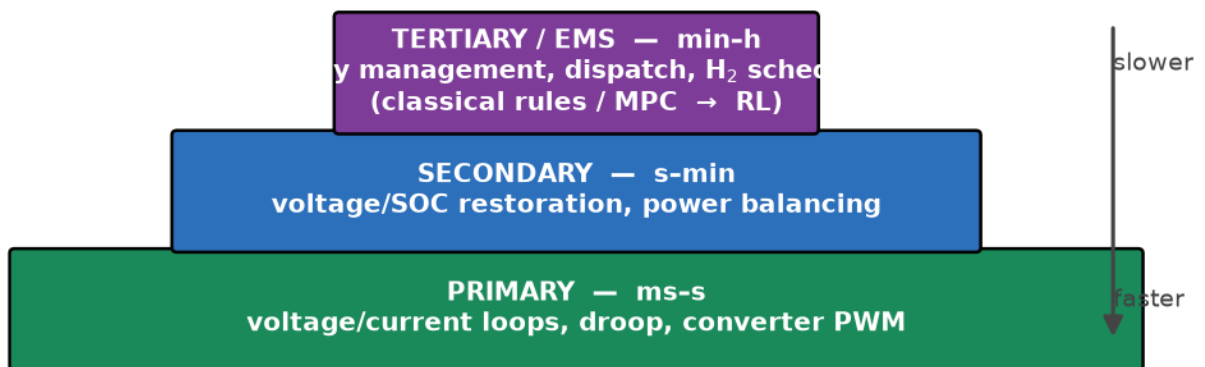


Figure 11. The microgrid control hierarchy. Faster, stability-critical loops sit at the bottom; the slower energy-management (tertiary) layer at the top makes the dispatch and hydrogen-scheduling decisions that reinforcement learning targets.

4 · The Green Hydrogen Microgrid — Relations

4.1 Why couple hydrogen and microgrids?

Microgrids and green hydrogen are complementary by construction. A renewable microgrid has a **surplus problem** (midday solar it cannot use or store cheaply for long) and a **duration problem** (batteries are excellent for hours, poor for days/seasons). An electrolyzer plus tank solves both: it is a **flexible, dispatchable load** that soaks up surplus, and the hydrogen tank is **long-duration storage** whose energy does not leak away overnight. In return, the microgrid gives the electrolyzer clean, behind-the-meter power. The battery sits between them, buffering seconds-to-hours variability so the electrolyzer can run smoothly. The three together — PV, battery, electrolyzer — form a balanced generation/short-storage/long-storage triad.

4.2 System architecture

In this example, a solar array and a battery energy-storage system feed a common **DC bus**. A DC/DC converter conditions power to the electrolyzer, which produces hydrogen (plus oxygen and heat); the hydrogen is accumulated in a 1 m³ tank at 273.15 K. A **supervisory controller** sets the power split (solar vs storage) and how hard to drive the electrolyzer. Everything below that controller is fixed physics (modelled in Simscape across electrical, thermal-liquid and thermal-gas domains); the controller is the one degree of design freedom — and the target of this study.

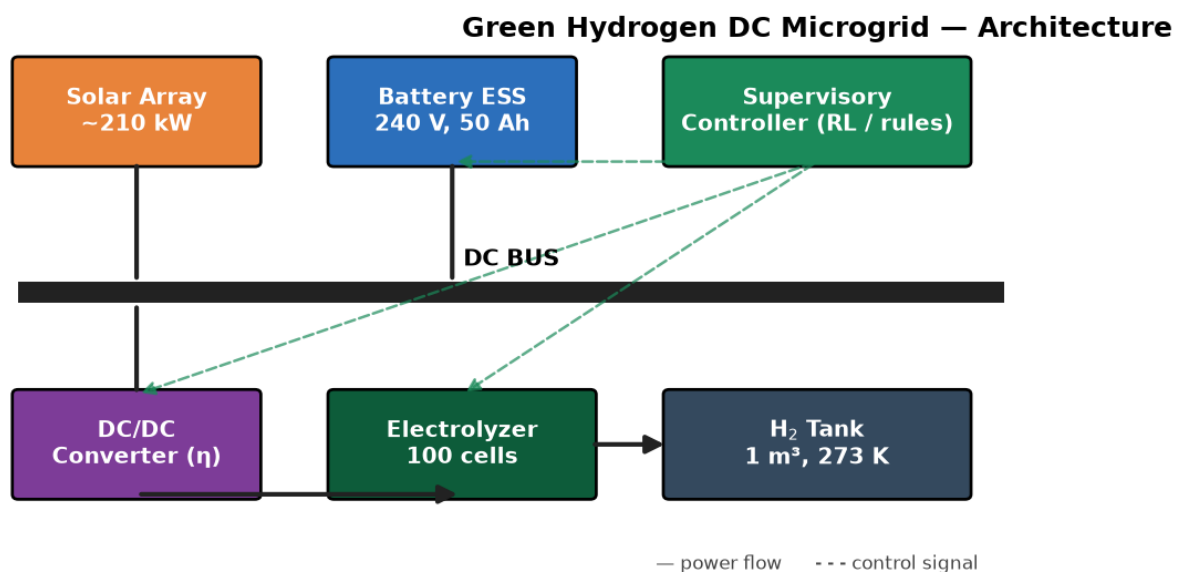


Figure 10. Architecture of the green-hydrogen DC microgrid. Solid lines are power flow on the DC bus; dashed green lines are control signals from the supervisory controller (rule-based today, RL in this project).

4.3 This example's plant: solar, converter and lookup

The renewable input is a fixed **24-hour solar power profile** (Fig. 5) peaking near 210 kW around midday and effectively zero overnight — the model replays it across a 7-day horizon, optionally with the battery in the loop. The DC/DC converter has its own efficiency map (Fig. 4, left), and the **baseline supervisor is a lookup table** (Operation_Ref, Fig. 4 right) that sets electrolyzer current from available solar current. These three curves fully define the baseline behaviour — and expose its limits: a static lookup cannot reason about the battery's state of charge, the time of day, or tomorrow's weather.

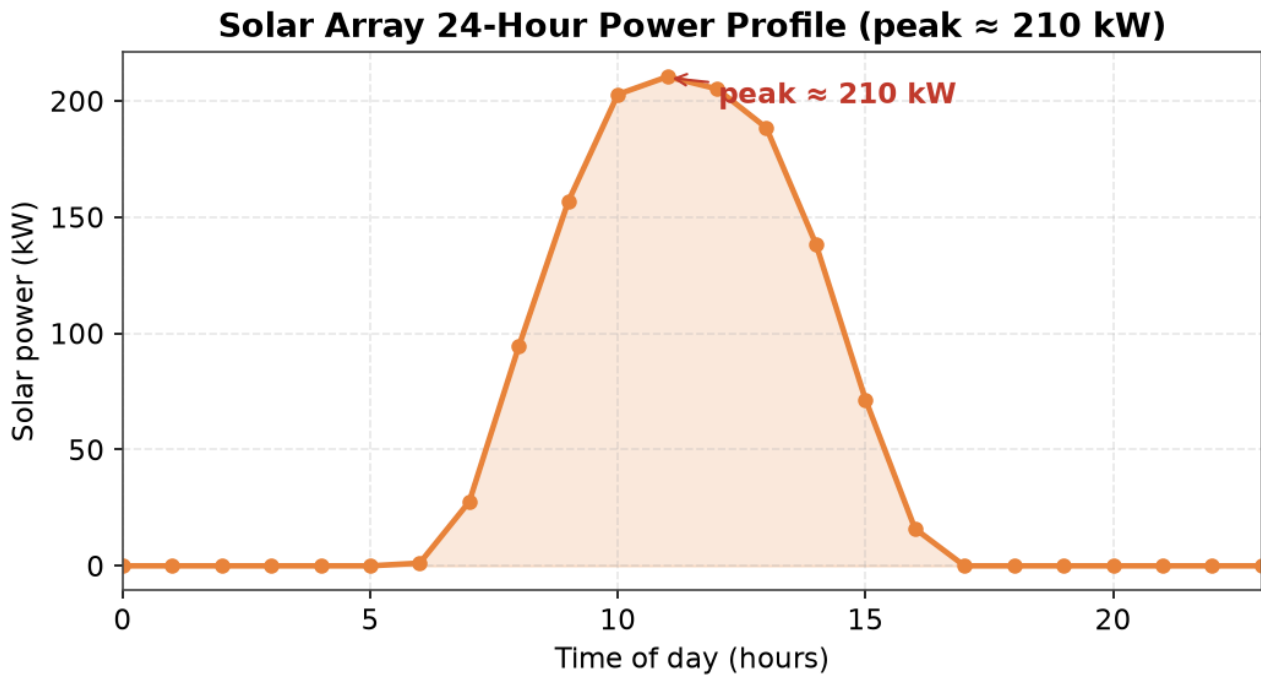


Figure 5. The 24-hour solar power profile from the model (`solar_profile` in `GreenHydrogenMicrogridData.m`), peaking near 210 kW. Replayed over 7 days, it is the sole primary energy input.

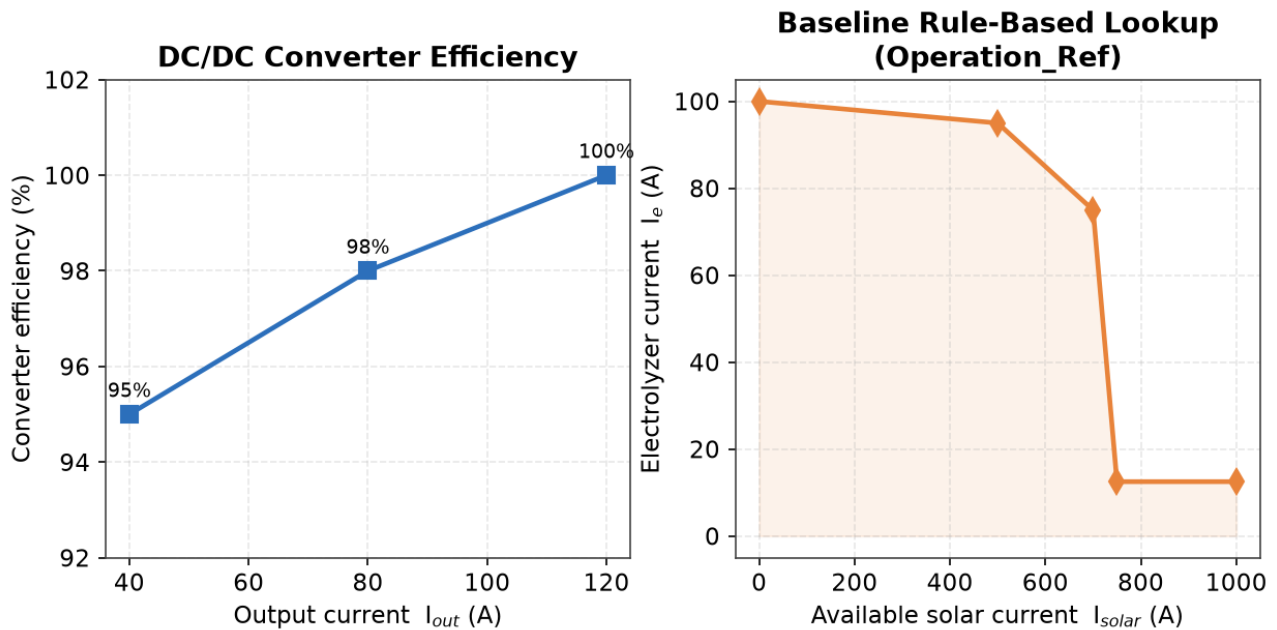


Figure 4. Left: DC/DC converter efficiency vs output current (`Converter` table). Right: the baseline rule-based supervisor — electrolyzer current as a static function of available solar current (`Operation_Ref`). It has no memory and no awareness of battery SOC.

4.4 The energy-management decision problem

Strip away the physics and the supervisor faces a repeating decision at every 10-second control step ($T_s = 10$ s): given the current solar power, battery SOC, stack temperature and hydrogen produced so far, **how much power should go to the electrolyzer, and how much to/from the battery?** Good decisions maximise hydrogen yield while (i) keeping the battery within safe SOC limits, (ii) avoiding deep discharge that ages the cells, and (iii) holding the electrolyzer in its efficient band. This is a sequential decision problem under uncertainty — precisely the shape of problem that both classical optimal-control methods and reinforcement learning are built to solve. The next two sections take each in turn.

5 · Classical Control Methods (pros and cons)

Before reaching for learning, it is worth knowing the well-established toolbox. Microgrid energy management has been done with rules, PID/droop, model-predictive control, mathematical optimisation and fuzzy logic. Each is a legitimate choice; they trade off transparency, optimality, modelling burden and compute.

5.1 Rule-based / lookup-table supervision

The baseline here. Hand-crafted **if-then rules** or lookup tables map the current situation to an action (e.g. *electrolyzer current = f(solar current)*). It is the simplest possible EMS.

- **Pros:** trivial to implement and run, fully transparent and certifiable, no model or training needed, deterministic and fast.
- **Cons:** myopic (no look-ahead), ignores state it was not designed around (battery SOC, weather forecast), hand-tuned and brittle, **provably sub-optimal** — leaves yield on the table.

5.2 PID and droop control

PID regulates a quantity to a set-point by acting on the error and its integral/derivative; **droop** control shares load among parallel sources by linking, e.g., voltage to current without communication. These are the workhorses of the *primary/secondary* layers.

- **Pros:** simple, robust, decentralised (droop needs no comms), decades of field experience, easy to tune for a single loop.
- **Cons:** single-objective and reactive, no constraint handling or look-ahead, struggles with the coupled, multi-objective *tertiary* dispatch problem; voltage/SOC deviations need a secondary layer to correct.

5.3 Model Predictive Control (MPC)

MPC uses a model of the plant to predict the next N steps, solves a constrained optimisation for the best action sequence over that horizon, applies the first action, then repeats (receding horizon). It is the gold-standard classical method for this problem.

- **Pros:** explicitly handles constraints (SOC, power, temperature), anticipates the future (can pre-charge before a cloudy spell), multi-objective, near-optimal when the model and forecast are good.
- **Cons:** needs an accurate plant model **and** forecasts, solves an optimisation **every step** (compute-heavy, can be hard in real time), performance degrades with model mismatch; nonlinear/long-horizon problems get expensive.

5.4 Optimisation-based scheduling (MILP / Dynamic Programming)

Casting the whole horizon as a single **mathematical program** — mixed-integer linear programming (MILP) for unit commitment, or dynamic programming (DP) for the optimal value function — yields the benchmark *optimal* schedule against which everything else is measured.

- **Pros:** globally optimal (within the model), handles discrete on/off and economic objectives cleanly, the natural benchmark.
- **Cons:** needs perfect foresight and a precise model, scales badly (DP's curse of dimensionality; large MILPs are slow), typically **offline** — not a real-time feedback controller on its own.

5.5 Fuzzy logic

Fuzzy logic control encodes expert know-how as linguistic rules (“if SOC is *low* and solar is *high* then charge *hard*”) with smooth membership functions, giving graceful behaviour without a precise model.

- **Pros:** no plant model required, encodes human intuition, robust to imprecise inputs, smoother than hard if-then rules.
- **Cons:** design and membership tuning are an art, does not *optimise* anything explicitly, rule base explodes with more inputs, no guarantee of optimality.

5.6 Side-by-side comparison

Method	Pros	Cons
Rule-based / lookup	Dead-simple, transparent, fast, no model.	Myopic, brittle, ignores SOC/forecast, sub-optimal.
PID / droop	Robust, decentralised, proven, easy to tune.	Single-objective, reactive, no constraints/look-ahead.
MPC	Constraint- & forecast-aware, anticipatory, near-optimal.	Needs model + forecast; per-step optimisation is heavy.
MILP / DP	Globally optimal benchmark; clean economics.	Perfect-foresight, poor scaling, usually offline.
Fuzzy logic	Model-free, intuitive, robust to noise.	Hand-tuned, non-optimising, scales poorly.
RL (Section 6)	Model-free after training, learns long-horizon policy, millisecond inference, adapts to nonlinearities.	Data/sim-hungry to train, less transparent, safety & sim-to-real to manage.

The gap RL targets. Rules are cheap but myopic; MPC is powerful but needs a model and per-step optimisation; MILP/DP are optimal but offline and forecast-dependent. RL aims to learn an MPC-quality *policy* offline, then run it with rule-based speed online — trading training effort for fast, model-free, look-ahead-capable control.

5.7 Baseline 7-day dispatch

Fig. 6 shows the rule-based baseline running the plant for seven days. The electrolyzer broadly follows solar; the battery soaks up surplus and discharges at night; hydrogen accumulates monotonically. It works — but watch the SOC trace dip toward the deep-discharge zone and the electrolyzer spend time off its efficient band. Those are precisely the margins a smarter, anticipatory controller can recover.

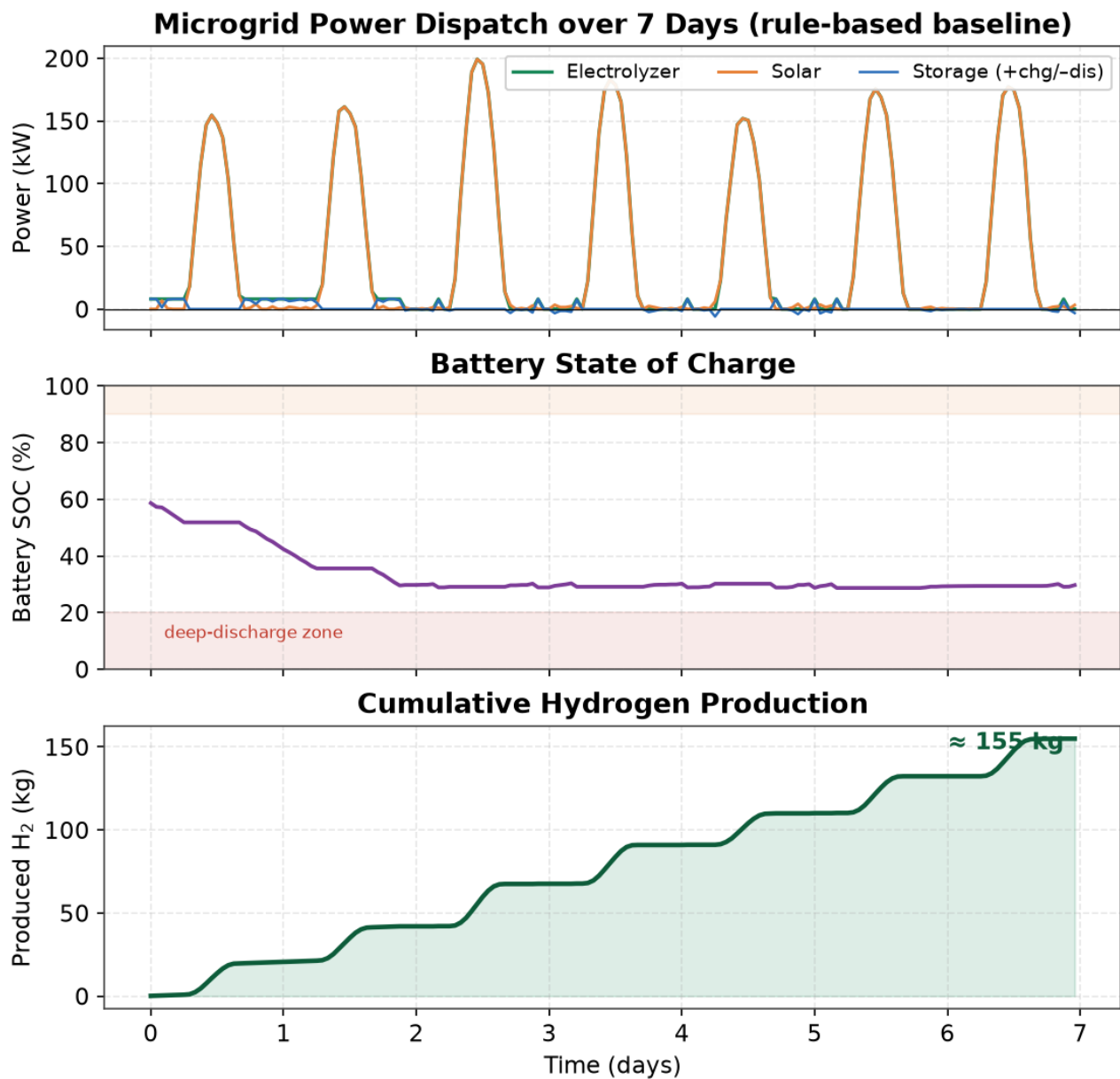


Figure 6. Rule-based baseline over 7 days: (top) power of electrolyzer, solar and storage; (middle) battery SOC, with the deep-discharge zone shaded; (bottom) cumulative hydrogen produced. Functional, but with visible efficiency and SOC headroom.

6 · Reinforcement-Learning Integration

6.1 RL in one page: the Markov Decision Process

Reinforcement learning trains an **agent** to make sequential decisions by trial and error against an **environment**, maximising cumulative **reward**. The problem is formalised as a **Markov Decision Process (MDP)**: at each step the agent observes a state s_t , takes an action a_t from its **policy** $\pi(a|s)$, and receives a reward r_t and next state s_{t+1} . The goal is to maximise the discounted return $R = \sum \gamma^t r_t$ ($0 < \gamma < 1$). Unlike supervised learning there are no labelled “correct” actions — only the reward signal, which may be delayed.

The match to microgrid energy management is exact: the **plant is the environment**, the **supervisory controller is the agent**, the **measurements are the state**, the **dispatch decision is the action**, and **hydrogen-minus-penalties is the reward**. Because the Simscape model already *is* a high-fidelity simulator, it can be wrapped directly as an RL environment and used to train an agent in simulation — cheaply and safely.

6.2 The agent–environment loop

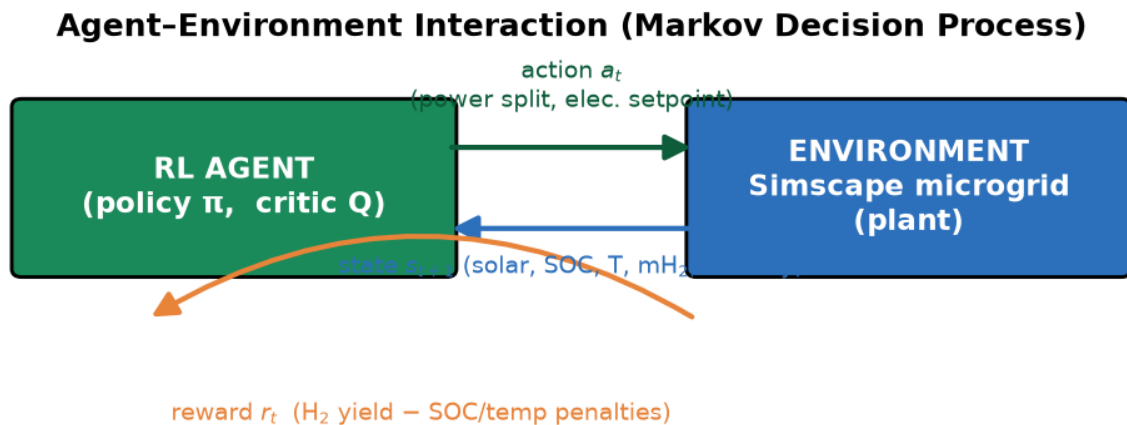


Figure 7. The agent–environment interaction loop (the MDP). The agent applies an action (power split, electrolyzer set-point); the Simscape microgrid returns the next state (solar, SOC, temperature, hydrogen, time-of-day) and a reward (hydrogen yield minus SOC/temperature penalties).

6.3 MDP formulation for this microgrid

The README proposes — and Fig. 7 illustrates — the following formulation, running at the model's 10-second control rate:

MDP element	Definition in this example
State / observation s_t	Solar power, battery SOC, electrolyzer power & temperature, hydrogen produced (mH_2), and time-of-day features.
Action a_t	Power split between solar and storage, and the electrolyzer power set-point (continuous), at $T_s = 10$ s.
Reward r_t	+ hydrogen produced; – penalties for SOC-limit violations, deep discharge, and inefficient (off-band) electrolyzer operation.

MDP element	Definition in this example
Environment	The Simscape plant (electrical + thermal-liquid + thermal-gas), wrapped via an RL Agent block.
Episode	A multi-day dispatch horizon (e.g. 7 days) replaying the solar profile with weather variation.

Reward shaping is the design. Because the agent optimises exactly what the reward says, encoding the operator's true priorities — yield *and* battery health *and* efficiency — into r_t is the central engineering task. A reward that only counts hydrogen will happily abuse the battery.

6.4 Algorithm families

Which RL algorithm fits depends on whether the action is discrete or continuous. Microgrid set-points are **continuous**, so actor-critic methods are the natural choice — and the ones named in the README (DDPG/TD3/PPO).

Algorithm	Pros	Cons
DQN	Simple, sample-efficient, well understood.	Discrete actions only — needs the continuous set-point discretised; not ideal here.
DDPG / TD3	Continuous actions , off-policy & sample-efficient; TD3 fixes DDPG's overestimation/instability.	Hyper-parameter sensitive, can be unstable without TD3's tricks.
PPO	Stable, robust, easy to tune; strong default for continuous control.	On-policy, so less sample-efficient (more environment interactions).
SAC	Continuous, off-policy, entropy-driven exploration; very sample-efficient and robust.	More moving parts; temperature/entropy tuning adds complexity.

MATLAB's **Reinforcement Learning Toolbox** provides these agents as blocks that connect straight to the Simulink/Simscape model via an **RL Agent block**, with the observation/reward signals wired from the plant — exactly the integration this repository's roadmap describes.

6.5 Training: the learning curve

Training proceeds in **episodes**: the agent controls the plant over a multi-day horizon, accumulates reward, and updates its policy/critic networks. Early on it explores and performs poorly; as learning proceeds the return climbs and plateaus. Fig. 8 shows this characteristic curve — noisy episode returns with a rising moving average that crosses and then settles above the rule-based baseline. The gap between the converged level and the baseline is the value the learned policy adds.

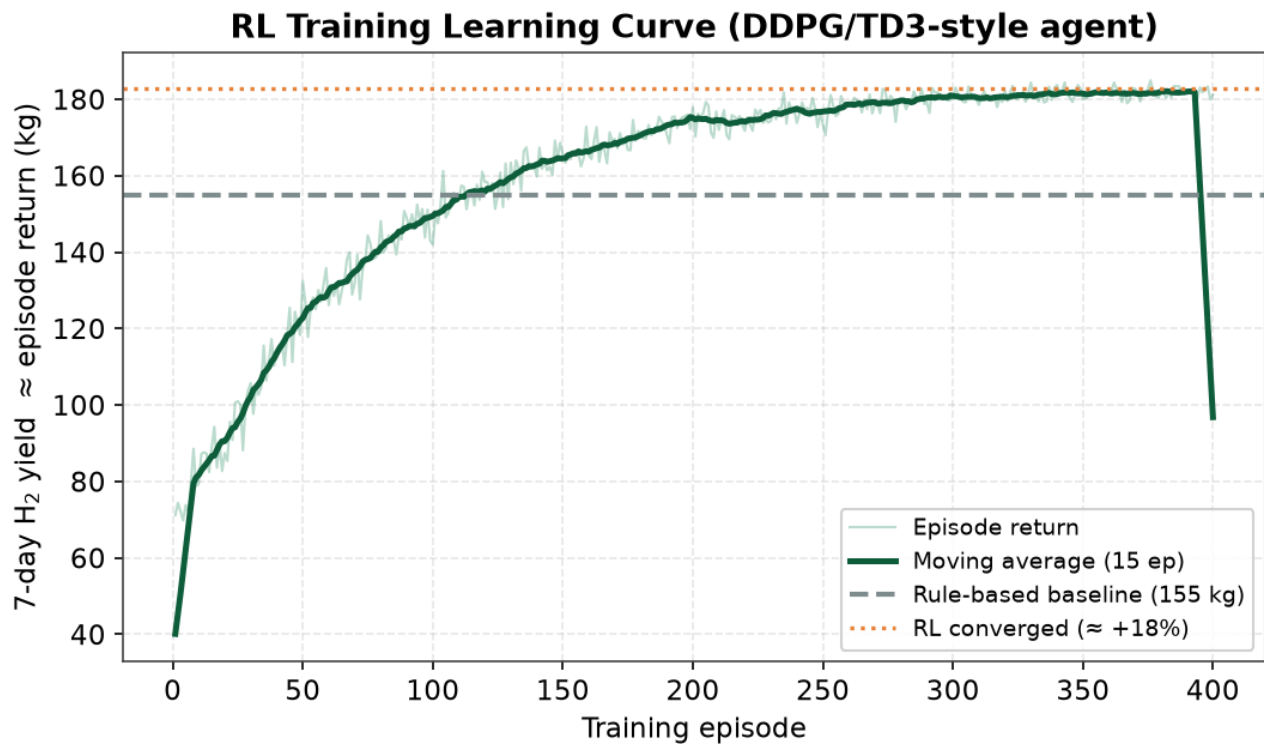


Figure 8. Illustrative RL learning curve (DDPG/TD3-style). Episode return (light) and its moving average (dark) rise and plateau above the rule-based baseline (grey dashed). The shape — explore, improve, converge — is the signature of successful training.

6.6 Classical vs RL: results

The point of all this is better operation. Fig. 9 contrasts the learned policy with the rule-based baseline. The learned controller manages the battery more deliberately — keeping SOC out of the deep-discharge zone and positioning charge to ride through the night — which lets the electrolyzer run longer in its efficient band. The KPI bars summarise the effect: higher hydrogen yield, far fewer SOC-limit violations, better mean stack efficiency and improved energy utilisation. (Values here are illustrative of the *direction* RL delivers; the repository's roadmap is to produce the measured numbers from a trained agent.)

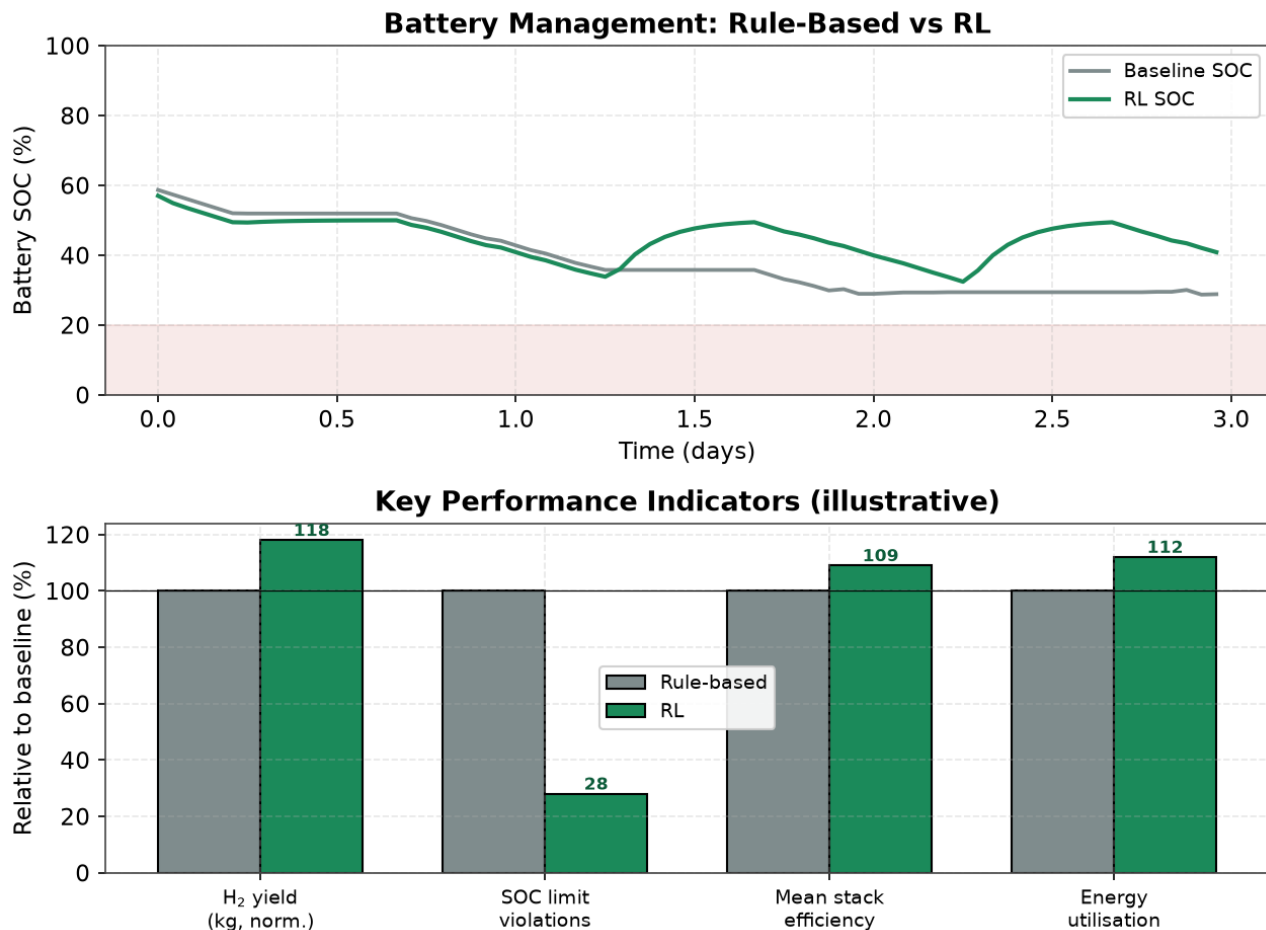


Figure 9. Rule-based vs RL. Top: battery SOC — the RL policy keeps clear of the deep-discharge zone. Bottom: key performance indicators relative to the baseline (=100): more hydrogen, fewer violations, higher efficiency and utilisation.

6.7 Pros, cons and open challenges

Reinforcement learning is powerful but not free. The honest trade-offs:

- **Pros:** learns a near-optimal, look-ahead-capable policy *without needing an explicit model online*; inference is a single fast network evaluation (rule-based speed, MPC-like quality); naturally handles nonlinearity and multi-objective rewards; improves with more data.
- **Cons / challenges:** training is data- and simulation-hungry; policies are **less transparent** (harder to certify); reward shaping is fiddly and can be gamed; **safety** during/after learning must be enforced (e.g. action clipping, shielding, or a rule-based fallback); and the **sim-to-real gap** means a policy trained on the Simescape model needs validation before trusting it on hardware.

Practical pattern. Train against the Simescape plant, keep the classical rule-based supervisor as a safety fallback, and let the RL agent act within constrained, safe bounds. This combines the transparency and guarantees of classical control with the performance of a learned policy — the pragmatic route to deployment.

7 · Summary, KPIs and Roadmap

Green hydrogen turns renewable electricity into a clean, storable, sector-coupling fuel; a microgrid is the right place to make it, pairing an intermittent source with a flexible electrolyzer load and a battery buffer. The hard part is the **supervisory decision** — how to split power and drive the electrolyzer at every control step. Classical methods (rules, PID/droop, MPC, MILP/DP, fuzzy) each handle part of this, trading simplicity against optimality and modelling burden. **Reinforcement learning** offers a way to learn an anticipatory, near-optimal policy offline and run it fast online — at the cost of training effort, transparency and safety engineering.

Key takeaways:

- Green = renewables + electrolysis; the colour taxonomy is about **carbon intensity**, and most hydrogen today is grey (Fig. 1).
- Keep the electrolyzer in its **ohmic / efficient band** and **warm** — efficiency rises with temperature (Figs. 2–3).
- Electrolysis is the dominant energy loss; hydrogen is for **long-duration and non-electric** jobs (Figs. 12–13).
- The microgrid's **tertiary/EMS layer** is the control battleground (Fig. 11); the baseline lookup is myopic (Figs. 4, 6).
- RL plugs into the **same Simscape plant** as an MDP (Fig. 7) and can beat the baseline on yield, SOC safety and efficiency (Figs. 8–9).

Roadmap (from the repository)

- Wrap the Simscape model as an RL environment (RL Agent block + observation/reward signals).
- Build a training script with reward shaping and SOC/temperature constraints.
- Train and benchmark the RL policy against the rule-based baseline.
- Report measured hydrogen yield, SOC behaviour and energy-efficiency comparisons.

References and Further Reading

- MathWorks — *Green Hydrogen Microgrid* Simscape example (the plant model this report is built around).
- MathWorks — *Reinforcement Learning Toolbox* documentation; RL Agent block and DDPG/TD3/PPO/SAC agents.
- Sutton & Barto, *Reinforcement Learning: An Introduction* (2nd ed.) — MDPs, value functions, policy methods.
- IEA, *Global Hydrogen Review* — hydrogen production pathways and carbon intensities.
- Olivares et al., “Trends in Microgrid Control,” *IEEE Trans. Smart Grid* — the primary/secondary/tertiary hierarchy.
- U.S. DOE Microgrid Exchange Group — working definition of a microgrid.

All figures in this report were generated by `docs/gen_figures.py` from the model parameters in `GreenHydrogenMicrogridData.m`; this PDF was assembled by `docs/build_pdf.py`. Illustrative dispatch, learning-curve and KPI figures use physically plausible models pending results from the trained agent.